

Wage Inequality, Upskilling, and Long-Run Growth:

A Three-Player Model with Linearly Approximated Risk Premia

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Abstract

We develop a tractable three-player model connecting wage inequality, worker upskilling decisions, and firm asset accumulation. A single parameter $\mu \in (0, \frac{1}{2})$ — the income share of the lower-ranked worker group — governs the entire wage distribution. Workers allocate income between upskilling investment and participation in a zero-sum displacement lottery. Risk preferences are captured through a linearly approximated Friedman-Savage utility, preserving the sign asymmetry between risk-seeking lower-income and risk-averse higher-income workers while keeping all equilibrium objects in closed form. The population mass ratio $W = i^*/(N - i^*)$ does not enter any individual worker’s budget or participation constraint, but enters the firm’s asset accumulation through the aggregate upskilling index $\Phi(\mu) = (W + 1)\kappa^{\min}(\mu)/R$, which is the population-weighted workforce contribution to the firm’s return on its wage bill. We address two questions: how do workers respond to wage inequality μ , and what constraints does μ place on long-term growth? We derive fully explicit steady-state equilibria and a complete set of signed comparative statics for all three players.

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1 Introduction

Wage inequality is central to debates about human capital investment, firm performance, and long-run economic growth. Standard growth models treat income distribution as either exogenous or absent, leaving unanswered the question of how a firm's wage policy feeds back into the skill investments of its workers and ultimately into the firm's own asset accumulation. This paper proposes a unified framework in which a single wage inequality parameter μ connects all three players — a lower-income worker, a higher-income worker, and a firm — and delivers closed-form answers to two sharply posed questions.

The first question is how workers respond to wage inequality μ . When the firm changes the wage split, it simultaneously alters each worker's income, upskilling investment, and lottery participation. The model delivers signed predictions for upskilling fractions, lottery stakes, net transfers, and risk premia as explicit functions of μ . A notable feature is the role of the population mass ratio W : it does not enter any individual worker's budget constraint or participation constraint, so individual upskilling fractions $\lambda_s^*(\mu)$ and the minimum upskilling floor $\kappa^{\min}(\mu)$ are independent of W . The population structure affects workers only through the lottery pool size and the net transfer direction, both of which depend on W through the prize mechanism. At the aggregate level, W amplifies the economy-wide human capital contribution through the upskilling index $\Phi(\mu) = (W + 1)\kappa^{\min}(\mu)/R$ — the quantity the firm internalises when setting wages.

The second question is what constraints μ places on long-term growth. The firm's asset accumulation is driven by $\Phi(\mu)$, net of depreciation and restructuring costs. Not all values of μ support positive growth: a viability condition must hold, depending on μ through the upskilling floor $\kappa_0 = \kappa^{\min}(\mu_0)$ and on W through $(W + 1)$. Conditional on viability, the firm's optimal μ^* is linear in the asset stock — wealthier firms enforce greater equality. Below the viability threshold, assets collapse regardless of initial conditions, constituting a poverty trap.

The model uses a linear approximation of the second derivative of the Friedman-Savage utility. This middle ground preserves the sign asymmetry between workers' risk premia — the lower-income worker has a negative risk premium (risk-seeking) and the higher-income worker a positive one (risk-averse) — while keeping all equilibrium objects in closed form. Crucially, risk preferences affect only the risk premia and not the upskilling or accumulation equilibrium, a clean separation that follows directly from the approximation structure.

2 Model Setup

2.1 Players, Incomes, and the Role of μ

There are three strategic players: worker L , worker H , and a firm. The firm holds asset stock A_t and distributes total income R to workers each period. The firm chooses the income-split parameter $\mu \in (0, \frac{1}{2})$:

$$M_L = R\mu, \quad M_H = R(1 - \mu) \quad (1)$$

Since $\mu < \frac{1}{2}$, worker H always earns more than worker L . Lower μ means greater wage inequality; higher μ means greater equality. The income ratio $M_H/M_L = (1 - \mu)/\mu$ is strictly decreasing in μ .

Definition 1 (Wage inequality parameter). $\mu \in (0, \frac{1}{2})$ is the income share of worker L , chosen by the firm each period. Greater wage equality corresponds to higher μ .

In the underlying N -worker displacement model with Pareto income ladder $m_i = x_0(1 - i/(N + 1))^{-1/\alpha}$ and Friedman-Savage class barrier at rank i^* , the two-player reduction yields:

$$\mu \approx \frac{\int_0^{i^*/N} x_0(1 - s)^{-1/\alpha} ds}{\int_0^1 x_0(1 - s)^{-1/\alpha} ds} \quad (2)$$

the lower hump's share of total representative income in the stationary distribution. In the macroeconomic model μ is treated as the firm's policy instrument.

2.2 The Population Mass Ratio W

The population mass ratio is:

$$W = \frac{i^*}{N - i^*} \geq 1 \quad (3)$$

where i^* is the class barrier rank. Under the Pareto distribution the lower group is generically more numerous, so $W \geq 1$.

The parameter W enters the model in two ways. First, it amplifies worker L 's aggregate lottery stake in the prize pool, reflecting that lower-rank workers generate more displacement events per period. Second, it weights the lower group's upskilling investment in the firm's aggregate upskilling index $\Phi(\mu) = (W + 1)\kappa^{\min}(\mu)/R$.

Importantly, W does *not* enter any individual worker's budget constraint or participation constraint. Each representative worker invests $\kappa^{\min}(\mu)$ regardless of W . The population weighting is an economy-level aggregation that the firm internalises, not an individual

incentive.

2.3 Worker Budget and Allocation

Each worker $s \in \{L, H\}$ allocates fraction $\lambda_s \in [0, 1]$ of income M_s to upskilling and the remainder to the lottery:

$$\kappa_s = \lambda_s M_s \quad (\text{upskilling}), \quad \ell_s = (1 - \lambda_s) M_s \quad (\text{lottery stake}) \quad (4)$$

Upskilling produces output $f(\kappa_s) = \kappa_s^\gamma$, $\gamma \in (0, 1)$, with diminishing returns. The certain base income after upskilling and human capital depreciation $\delta_w > 0$:

$$b_s(\lambda_s) = \kappa_s + \kappa_s^\gamma - \delta_w \quad (5)$$

2.4 Zero-Sum Displacement Lottery

The lottery represents the displacement mechanism from the N -worker model. Workers contribute stakes to a common pool and one side wins the entire pool each period. The pool is:

$$\Lambda = W(1 - \lambda_L)M_L + (1 - \lambda_H)M_H = W\ell_L + \ell_H \quad (6)$$

The factor W amplifies the lower group's contribution, consistent with their greater population mass. Each worker wins the pool with probability $\frac{1}{2}$. The outcome incomes are:

$$M_L^{\text{win}} = b_L + (1 - \lambda_H)M_H \quad (7)$$

$$M_L^{\text{lose}} = b_L - W(1 - \lambda_L)M_L \quad (8)$$

$$M_H^{\text{win}} = b_H + W(1 - \lambda_L)M_L \quad (9)$$

$$M_H^{\text{lose}} = b_H - (1 - \lambda_H)M_H \quad (10)$$

The lottery is strictly zero-sum: only upskilling changes the aggregate income.

2.5 Participation Constraint and Upskilling Floor

Each worker must maintain base income above a subsistence floor $\bar{b}(\mu) = \bar{b}_0 R\mu$, proportional to the lower worker's wage:

$$b_s(\lambda_s) \geq \bar{b}(\mu) \iff \kappa_s + \kappa_s^\gamma \geq \bar{b}_0 R\mu + \delta_w \quad (11)$$

The minimum upskilling investment satisfying this constraint, $\kappa^{\min}(\mu)$, solves:

$$\kappa^{\min} + (\kappa^{\min})^\gamma = \bar{b}_0 R \mu + \delta_w \quad (12)$$

For $\gamma = 2$, adopted throughout for closed-form solutions:

$$\kappa^{\min}(\mu) = \frac{-1 + \sqrt{1 + 4(\bar{b}_0 R \mu + \delta_w)}}{2} \quad (13)$$

This is strictly increasing and concave in μ , with sensitivity:

$$\frac{d\kappa^{\min}}{d\mu} \equiv \Psi(\mu) = \frac{\bar{b}_0 R}{1 + 2\kappa^{\min}(\mu)} > 0 \quad (14)$$

The same sensitivity $\Psi(\mu)$ appears in both the worker upskilling response (Section 4) and the firm's first-order condition (Section 5), linking the two sides of the model.

2.6 The Aggregate Upskilling Index

The firm internalises the population-weighted aggregate upskilling contribution of its workforce through the upskilling index:

$$\Phi(\mu) = \frac{(W + 1)\kappa^{\min}(\mu)}{R} \quad (15)$$

This is the aggregate human capital return per unit of total income. The factor $(W + 1)$ reflects that there are W lower-rank workers for every upper-rank worker, so the aggregate upskilling output per representative pair is $(W + 1)\kappa^{\min}(\mu)$ rather than $2\kappa^{\min}(\mu)$. The derivative:

$$\frac{d\Phi}{d\mu} = \frac{(W + 1)\Psi(\mu)}{R} > 0, \quad \frac{d^2\Phi}{d\mu^2} = \frac{(W + 1)}{R} \frac{d\Psi}{d\mu} < 0 \quad (16)$$

The upskilling index is strictly increasing and strictly concave in μ : returns to equality are positive but diminishing. A larger W raises both the level of Φ and its sensitivity to μ , but does not affect the workers' individual $\kappa^{\min}(\mu)$ or $\lambda_s^*(\mu)$.

3 Worker Preferences and Linearly Approximated Risk Premia

3.1 Friedman-Savage Utility

Worker utility uses a cubic Friedman-Savage specification with inflection point m^* :

$$u(m) = m - \frac{\alpha_u}{2}(m - m^*)^2 + \frac{\beta_u}{3}(m - m^*)^3 \quad (17)$$

where $\alpha_u, \beta_u > 0$. The second derivative $u''(m) = -\alpha_u + 2\beta_u(m - m^*)$ is negative for $m > m^*$ (concave, risk-averse) and positive for $m < m^*$ (convex, risk-seeking). Worker L with $M_L = R\mu < m^*$ is risk-seeking; worker H with $M_H = R(1 - \mu) > m^*$ is risk-averse.

3.2 The Linear Approximation of u''

We linearise the second derivative around the subsistence income $\bar{b}(\mu)$:

$$u''(m) \approx u''(\bar{b}) + 2\beta_u(m - \bar{b}) \quad (18)$$

Since $u'''(m) = 2\beta_u$ is constant under the cubic specification, equation (18) is exact for u'' itself. The approximation enters when we replace $u'(m)$ by $u'(\bar{b})$ in the Arrow-Pratt coefficient:

$$A(m) \approx \tilde{A}(m) = \frac{-u''(m)}{u'(\bar{b})} = \frac{\alpha_u - 2\beta_u(m - m^*)}{u'(\bar{b})} \quad (19)$$

This is valid when u' does not vary substantially over the lottery outcome range, holding when prizes are small relative to $|m^* - \bar{b}|$.

3.3 Linearly Approximated Risk Premia

We use the following compact notation throughout:

$$a(\mu) = R\mu - \kappa^{\min}(\mu) \quad (\text{worker } L \text{ lottery stake}) \quad (20)$$

$$c(\mu) = R(1 - \mu) - \kappa^{\min}(\mu) \quad (\text{worker } H \text{ lottery stake}) \quad (21)$$

$$\Pi(\mu) = Wa(\mu) + c(\mu) = \Lambda^* \quad (\text{total lottery spread}) \quad (22)$$

$$\epsilon_L(\mu) = \frac{c(\mu) - Wa(\mu)}{2} \quad (\text{net expected transfer to } L) \quad (23)$$

$$\epsilon_H(\mu) = -\epsilon_L(\mu) \quad (\text{net expected transfer to } H) \quad (24)$$

The linearly approximated risk premium for worker s is:

$$\tilde{\rho}_s = \frac{\Pi(\mu)^2}{8} \cdot \frac{\alpha_u - 2\beta_u D_s}{u'(\bar{b})} \quad (25)$$

where $D_s \equiv \bar{b}(\mu) - m^* + \epsilon_s$ is the deviation of worker s 's expected lottery outcome from the inflection point. Since $D_L < 0$ and $D_H > 0$:

$$\tilde{\rho}_L < 0 \quad (\text{risk-seeking}), \quad \tilde{\rho}_H > 0 \quad (\text{risk-averse}) \quad (26)$$

The net social risk premium collapses to:

$$\tilde{\rho}_L + \tilde{\rho}_H = -\frac{\Pi(\mu)^2}{4} \cdot A(\bar{b}) \quad (27)$$

The lottery is socially desirable if and only if $A(\bar{b}) < 0$, i.e. $\bar{b}(\mu) < m^*$. Since $\bar{b}(\mu) = \bar{b}_0 R \mu$ rises with μ , there exists a critical threshold:

$$\mu^\dagger = \frac{m^*}{\bar{b}_0 R} \quad (28)$$

above which the lottery mechanism becomes socially costly. This threshold is independent of W : population structure does not affect the social desirability of the displacement mechanism.

3.4 Equilibrium Utility

The worker's expected utility, expanded to second order under the linear approximation, is:

$$V_s \approx \underbrace{u(\bar{b})}_{\text{subsistence value}} + \underbrace{\epsilon_s \cdot u'(\bar{b})}_{\text{expected transfer}} + \underbrace{\tilde{\rho}_s}_{\text{risk premium}} \quad (29)$$

The utility parameters (α_u, β_u, m^*) enter only through $\tilde{\rho}_s$, not through the upskilling allocation or asset accumulation. This clean separation is the key property of the linear approximation and holds exactly at second order.

4 Question 1: Worker Responses to Wage Inequality

μ

4.1 Optimal Worker Strategies

Since the participation constraint $b_s \geq \bar{b}$ binds at equilibrium, the optimal upskilling fractions are:

$$\lambda_L^*(\mu) = \frac{\kappa^{\min}(\mu)}{R\mu}, \quad \lambda_H^*(\mu) = \frac{\kappa^{\min}(\mu)}{R(1-\mu)} \quad (30)$$

Both are explicit functions of μ through equation (13), and neither depends on W . Since $R\mu < R(1-\mu)$:

$$\lambda_L^*(\mu) > \lambda_H^*(\mu) \quad \forall \mu \in (0, \frac{1}{2}) \quad (31)$$

The lower-income worker always invests a larger fraction of income in upskilling because the same absolute floor $\kappa^{\min}$ represents a larger share of their smaller budget. As $\mu \rightarrow \frac{1}{2}$ the fractions converge; as $\mu \rightarrow 0$ the gap widens without bound.

4.2 Upskilling Responses to a Change in μ

Differentiating equation (30) with respect to μ :

$$\frac{d\lambda_L^*}{d\mu} = \frac{\Psi(\mu) - \kappa^{\min}(\mu)/\mu}{R\mu} \quad (32)$$

$$\frac{d\lambda_H^*}{d\mu} = \frac{\Psi(\mu) + \kappa^{\min}(\mu)/(1-\mu)}{R(1-\mu)} > 0 \quad \text{always} \quad (33)$$

Worker H always increases their upskilling fraction when equality rises because both terms in the numerator are positive. For worker L , the sign depends on whether $\Psi(\mu) > \kappa^{\min}(\mu)/\mu$, which holds when the participation constraint is tight relative to current upskilling — the typical case for moderate inequality.

The total upskilling investment $K^*(\mu)$ and its response:

$$K^*(\mu) = (W+1)\kappa^{\min}(\mu), \quad \frac{dK^*}{d\mu} = (W+1)\Psi(\mu) > 0 \quad (34)$$

Greater wage equality raises total upskilling investment. The response is amplified by $(W+1)$: a larger lower-rank population magnifies the economy-wide human capital response to any given change in wage policy. This amplification is absent from the individual fractions $\lambda_s^*(\mu)$, which do not depend on W , and emerges only at the aggregate level when population weighting is applied.

Proposition 1 (Upskilling and population structure). *At the constrained equilibrium:*

1. *Individual upskilling fractions $\lambda_L^*(\mu)$ and $\lambda_H^*(\mu)$, and the upskilling floor $\kappa^{\min}(\mu)$, are all independent of W . The population mass ratio does not affect individual worker investment decisions.*
2. *Total upskilling $K^*(\mu) = (W+1)\kappa^{\min}(\mu)$ is increasing in both μ and W . A larger lower-rank population amplifies the economy-wide human capital base for any given wage policy.*
3. *The same amplification factor $(W+1)$ appears in the upskilling index $\Phi(\mu) = K^*(\mu)/R$ and the firm's optimal μ^* -coefficient $\Gamma \propto (W+1)$, ensuring that firm behaviour and aggregate upskilling are consistently linked.*

4.3 Lottery Stakes and Net Transfer

The lottery stakes and their responses to μ :

$$a(\mu) = R\mu - \kappa^{\min}(\mu), \quad c(\mu) = R(1-\mu) - \kappa^{\min}(\mu) \quad (35)$$

$$\frac{da}{d\mu} = R - \Psi(\mu) > 0, \quad \frac{dc}{d\mu} = -R - \Psi(\mu) < 0 \quad (36)$$

As μ rises, worker L 's stake increases (income rises faster than the upskilling floor) while worker H 's falls (income falls and floor rises). The net expected transfer to worker L :

$$\epsilon_L(\mu) = \frac{c(\mu) - Wa(\mu)}{2} = \frac{R(1 - \mu) - WR\mu - (1 - W)\kappa^{\min}(\mu)}{2} \quad (37)$$

is decreasing in μ when $W > 1$: higher equality reduces worker L 's net lottery receipt because their growing stake means they contribute proportionally more to the pool. The transfer reversal threshold $W^\dagger = c(\mu)/a(\mu)$ gives the value of W above which worker L becomes a net contributor to the pool despite being the lower-income worker.

4.4 Risk Premium Responses to μ

As μ rises, $\bar{b}(\mu) = \bar{b}_0 R\mu$ increases, moving both workers toward the inflection m^* . Worker L 's D_L moves toward zero from below, so $|\tilde{\rho}_L|$ falls — risk-seeking diminishes as worker L becomes richer. Worker H 's D_H moves further above zero, so $\tilde{\rho}_H$ rises — risk-aversion strengthens as worker H 's relative advantage grows. The net social premium changes sign at $\mu = \mu^\dagger$, above which the lottery displacement mechanism imposes a net social cost.

Proposition 2 (Worker responses to wage equality). *Under the linear approximation of u'' :*

1. *Worker H always raises λ_H^* when μ increases. Worker L raises λ_L^* when the participation constraint is sufficiently tight.*
2. *Total upskilling $K^*(\mu) = (W + 1)\kappa^{\min}(\mu)$ is increasing in μ and in W .*
3. *Worker L 's risk-seeking premium magnitude $|\tilde{\rho}_L|$ falls with μ ; worker H 's risk-aversion premium $\tilde{\rho}_H$ rises with μ .*
4. *A threshold $\mu^\dagger = m^*/(\bar{b}_0 R)$, independent of W , marks the point above which the lottery is socially costly.*

4.5 Summary of Worker Responses

Table 1 collects the signed responses of all worker-side equilibrium objects to an increase in wage equality.

Table 1: Worker responses to an increase in wage equality ($\mu \uparrow$)

Object	Symbol	$\mu \uparrow$	Mechanism
Worker L income	M_L	+	Direct
Worker H income	M_H	−	Direct
Upskilling floor	$\kappa^{\min}$	+	Rising subsistence $\bar{b}_0 R \mu$
Worker L fraction	λ_L^*	+(conditional)	eq. (32), W -independent
Worker H fraction	λ_H^*	+(always)	Both channels positive, W -independent
Total upskilling	$K^* = (W+1)\kappa^{\min}$	+	Floor rises; amplified by W
Worker L stake	a	+	Income rises faster than floor
Worker H stake	c	−	Income falls and floor rises
Lottery pool	Λ^*	+(conditional)	Income vs. floor effect
Net transfer to L	ϵ_L^*	−	Larger L stake hurts L
L risk premium	$ \tilde{\rho}_L $	−	L closer to inflection
H risk premium	$\tilde{\rho}_H$	+	H further from inflection
Net social premium	$\tilde{\rho}_L + \tilde{\rho}_H$	sign flip at $\mu^\dagger$	$\bar{b}$ crosses m^*

Note: λ_L^* , λ_H^* , and $\kappa^{\min}$ are all independent of W . The W -dependence of K^* arises entirely from the population weighting $(W+1)$, not from any change in individual investment.

5 Question 2: μ Constraints on Long-Term Growth

5.1 The Firm's Asset Accumulation Problem

The firm accumulates assets A_t according to:

$$A_{t+1} = [(1 - \pi - \delta_f) + \pi \Phi(\mu_t)] A_t - K(\mu_t) - C(A_t) \quad (38)$$

where:

$$\Phi(\mu_t) = \frac{(W+1)\kappa^{\min}(\mu_t)}{R} \quad (39)$$

$$K(\mu_t) = k|\mu_t - \mu_0|^\eta, \quad k, \eta > 0 \quad (40)$$

$$C(A_t) = c_f A_t^\xi, \quad c_f, \xi > 0 \quad (41)$$

The wage share $\pi \in (0, 1)$ determines what fraction of assets is paid as wages. The upskilling index $\Phi(\mu_t)$ converts the wage bill into a productive return: wages paid at split μ_t induce upskilling investment $(W+1)\kappa^{\min}(\mu_t)$ across the workforce, which feeds back into asset growth at rate π . The restructuring cost $K(\mu_t) = k|\mu_t - \mu_0|^\eta$ is zero at the market reference μ_0 and rises convexly with deviation. The maintenance cost $C(A_t) = c_f A_t^\xi$ is the running cost of a larger asset base.

5.2 Stackelberg Timing

The firm solves a Stackelberg problem within each period. It moves first by choosing μ_t , anticipating workers' best responses $\lambda_s^*(\mu_t)$. By substituting these best responses into $\Phi(\mu_t)$ before optimising, the problem reduces to a single-variable maximisation with no circularity:

$$\mu_t^* = \arg \max_{\mu_t} \left\{ \left[(1 - \pi - \delta_f) + \frac{\pi(W + 1)\kappa^{\min}(\mu_t)}{R} \right] A_t - k|\mu_t - \mu_0|^\eta - c_f A_t^\xi \right\} \quad (42)$$

The workers' choices are substituted out, leaving $\Phi(\mu_t)$ as a known function of μ_t alone. Cross-period feedback (A_{t+1} informing μ_{t+1}^*) is handled by standard dynamic programming and introduces no within-period circularity.

5.3 Firm's First-Order Condition and Optimal μ^*

Taking the derivative of (42) with respect to μ_t and setting to zero, with $\eta = 2$ for closed form and Ψ evaluated at the reference $\Psi_0 \equiv \Psi(\mu_0)$:

$$\underbrace{\frac{\pi(W + 1)\Psi_0}{R} \cdot A_t}_{\text{marginal gain: aggregate upskilling return}} = \underbrace{2k(\mu_t^* - \mu_0)}_{\text{marginal cost: restructuring}} \quad (43)$$

The left-hand side is the marginal gain: raising μ_t^* increases $\kappa^{\min}$ by Ψ_0 , total upskilling by $(W + 1)\Psi_0$, and asset returns by $\pi(W + 1)\Psi_0 A_t / R$. The right-hand side is the marginal restructuring cost. Solving:

$$\boxed{\mu_t^* = \mu_0 + \Gamma A_t, \quad \Gamma \equiv \frac{\pi(W + 1)\Psi_0}{2kR}} \quad (44)$$

The coefficient Γ is proportional to $(W + 1)$: the firm's incentive to equalise wages is driven by the population-weighted aggregate upskilling return, not by any individual worker response. A larger lower-rank group makes equality more valuable to the firm because it amplifies the aggregate human capital payoff.

Proposition 3 (Firm's optimal wage inequality). *The optimal income split satisfies:*

1. $\mu_t^* = \mu_0 + \Gamma A_t$: optimal equality is linear in the asset stock.
2. $\mu_t^* > \mu_0$ for all $A_t > 0$: the firm always chooses greater equality than the market reference.
3. $\partial \mu_t^* / \partial A_t = \Gamma > 0$: wealthier firms enforce greater equality.
4. $\partial \mu_t^* / \partial W > 0$: a larger lower-rank population pushes the firm toward equality through

$(W + 1)$ in Γ .

5. $\partial\mu_t^*/\partial k < 0$: higher restructuring costs reduce optimal deviation from μ_0 .

5.4 Three Constraints of μ on Long-Term Growth

The asset growth rate is:

$$g(\mu, A) = \pi\Phi(\mu) - \delta_f - c_f - \frac{k(\mu - \mu_0)^2}{A} \quad (45)$$

Three distinct constraints on μ emerge.

Constraint 1 — Viability floor. Setting $\mu = \mu_0$ and $A \rightarrow \infty$ gives the maximum achievable growth rate $g^{\max} = \pi\Phi(\mu_0) - \delta_f - c_f$. A positive steady state requires:

$$\boxed{\pi(W + 1)\kappa_0 > R(\delta_f + c_f)} \quad (46)$$

where $\kappa_0 = \kappa^{\min}(\mu_0)$. When this fails, no wage policy can generate positive growth regardless of μ . The condition depends on W through $(W + 1)$: a larger lower-rank population makes viability easier to satisfy. Below this threshold the economy has no positive steady state and assets collapse — a poverty trap.

Constraint 2 — Restructuring upper bound. For the firm to benefit from $\mu_t > \mu_0$:

$$\pi[\Phi(\mu_t) - \Phi(\mu_0)]A_t > k(\mu_t - \mu_0)^2 \quad (47)$$

This is an asset-dependent upper bound on feasible μ_t . Richer firms can sustain more redistribution. As $A_t \rightarrow A^*$, this constraint binds exactly at $\mu_t = \mu_t^*$ from equation (44), confirming the interior optimum.

Constraint 3 — Hard feasibility ceiling. Regardless of asset level, $\mu < \frac{1}{2}$: worker H must always earn more than worker L . As $\mu \rightarrow \frac{1}{2}$ the upskilling gap $\lambda_L^* - \lambda_H^*$ closes and the lottery spread $\Pi(\mu) \rightarrow 0$, eliminating the displacement mechanism.

5.5 Closed-Form Steady-State Equilibrium

Setting $A_{t+1} = A_t = A^*$ and using $\mu^* - \mu_0 = \Gamma A^*$:

$$A^*[\pi\Phi^* - \delta_f - c_f] = k\Gamma^2 A^{*2} \quad (48)$$

Dividing by $A^* > 0$ and evaluating Φ^* at reference κ_0 :

$$\boxed{A^* = \frac{\pi(W + 1)\kappa_0/R - \delta_f - c_f}{k\Gamma^2}} \quad (49)$$

The full steady state is then computed sequentially. Define three derived constants from primitives:

$$\kappa_0 = \frac{-1 + \sqrt{1 + 4(\bar{b}_0 R \mu_0 + \delta_w)}}{2} \quad (50)$$

$$\Psi_0 = \frac{\bar{b}_0 R}{1 + 2\kappa_0} \quad (51)$$

$$\Gamma = \frac{\pi(W + 1)\Psi_0}{2kR} \quad (52)$$

Then all equilibrium objects follow without implicit equations:

$$\text{Firm:} \quad A^* = \frac{\pi(W + 1)\kappa_0/R - \delta_f - c_f}{k\Gamma^2} \quad (53)$$

$$\mu^* = \mu_0 + \Gamma A^* \quad (54)$$

$$\text{Workers:} \quad \kappa^* = \frac{-1 + \sqrt{1 + 4(\bar{b}_0 R \mu^* + \delta_w)}}{2} \quad (55)$$

$$\lambda_L^* = \frac{\kappa^*}{R\mu^*}, \quad \lambda_H^* = \frac{\kappa^*}{R(1 - \mu^*)} \quad (56)$$

$$K^* = (W + 1)\kappa^*, \quad a^* = R\mu^* - \kappa^*, \quad c^* = R(1 - \mu^*) - \kappa^* \quad (57)$$

$$\Lambda^* = W a^* + c^*, \quad \epsilon_L^* = \frac{c^* - W a^*}{2} \quad (58)$$

$$\text{Risk premia:} \quad \tilde{\rho}_s^* = \frac{\Lambda^{*2}}{8} \cdot \frac{\alpha_u - 2\beta_u D_s^*}{u'(\bar{b}^*)} \quad (59)$$

Every object is an explicit function of the twelve primitives $(R, \mu_0, W, \bar{b}_0, \delta_w, \pi, \delta_f, k, c_f, \alpha_u, \beta_u, m^*)$.

5.6 The Endogenous Kuznets Relationship

At the steady state, $\mu^* = \mu_0 + \Gamma A^*$, so wealthier economies operate at a more equal wage split. Along the transitional path, as A_t grows toward A^* , $\mu_t^* = \mu_0 + \Gamma A_t$ rises monotonically. The Gini coefficient of income (proxied by $1 - 2\mu^*$) falls as the economy grows: an endogenous Kuznets relationship in which greater wealth generates greater equality through the firm's optimal wage policy. This relationship is steeper when W is larger, because the aggregate upskilling amplification makes equality more valuable to richer firms.

6 Full Comparative Statics

Table 2 presents the complete signed comparative statics of the steady-state equilibrium. The W row illustrates Proposition 1 directly: λ_L^* and λ_H^* have ambiguous or negative

responses to W (reflecting that individual fractions do not have W in their formula, but respond indirectly through the higher μ^* that W induces), while K^* rises with W because $(W + 1)$ enters it directly.

Table 2: Steady-state comparative statics for all three players

Shock ($\uparrow$)	∂A^*	$\partial \mu^*$	$\partial \lambda_L^*$	$\partial \lambda_H^*$	∂K^*	$\partial \Lambda^*$	$\partial(\tilde{\rho}_L^* + \tilde{\rho}_H^*)$
$W \uparrow$	+	+	amb.	—	+	+	+
$\pi \uparrow$	+	+	+	+	+	+	+
$k \uparrow$	—	—	—	—	—	—	—
$\bar{b}_0 \uparrow$	amb.	+	+	—	+	amb.	—
$\delta_f \uparrow$	—	—	—	—	—	—	—
$\delta_w \uparrow$	+	+	+	amb.	+	+	+
$\mu_0 \uparrow$	+	+	amb.	—	+	+	—
$R \uparrow$	amb.	amb.	—	—	amb.	amb.	amb.
$\alpha_u \uparrow$	—	—	—	—	—	—	—
$\beta_u \uparrow$	—	—	—	—	—	—	+
$m^* \uparrow$	—	—	—	—	—	—	+

Notes: “amb.” = ambiguous sign depending on parameter magnitudes. “—” = parameter affects risk premia only; does not enter the allocation or accumulation equations.

$K^* = (W + 1)\kappa^{\min}(\mu^*)$ is the aggregate population-weighted upskilling total.

Several comparative statics merit comment. The wage share π has the strongest uniformly positive effect, raising all objects except possibly ϵ_L^* . Restructuring cost k uniformly reduces everything, as it discourages the firm from departing from μ_0 . Worker depreciation δ_w raises A^* and μ^* by raising κ_0 , which strengthens the upskilling return in the viability condition. Utility parameters (α_u, β_u, m^*) affect only the risk premia and not the allocation or accumulation objects, confirming the clean separation of the linear approximation.

For worker H ’s upskilling fraction, the response to W is negative because higher W raises μ^* , which raises $R(1 - \mu^*)$ in the denominator of $\lambda_H^* = \kappa^*/R(1 - \mu^*)$ by more than it raises κ^* in the numerator. Worker L ’s response to W is ambiguous for the same reason as $d\lambda_L^*/d\mu$: the numerator and denominator effects partially cancel.

7 Conclusion

We have developed a tractable three-player model in which wage inequality μ is the central instrument connecting worker behaviour and firm growth, and in which the population mass ratio W enters aggregate upskilling through the index $\Phi(\mu) = (W + 1)\kappa^{\min}(\mu)/R$ without appearing in any individual worker’s constraints.

On the first question — worker responses to μ — the main findings are: worker H always raises their upskilling fraction when equality rises; total upskilling $K^*(\mu) = (W + 1)\kappa^{\min}(\mu)$ rises with both μ and W , with the amplification factor $(W + 1)$ determined entirely by population structure and absent from the individual fractions $\lambda_s^*(\mu)$; worker L 's risk-seeking premium falls in magnitude as μ rises; a threshold $\mu^\dagger = m^*/(\bar{b}_0 R)$ marks the point above which the lottery displacement mechanism is socially costly.

On the second question — μ constraints on growth — the main findings are: the viability condition $\pi(W + 1)\kappa_0 > R(\delta_f + c_f)$ is the hard floor, with W making viability easier to satisfy; the optimal $\mu^* = \mu_0 + \Gamma A^*$ is linear in asset wealth, generating an endogenous Kuznets relationship; the steady-state asset level A^* is fully explicit and rises with W , π , and δ_w , and falls with k and δ_f .

The key structural insight unifying both answers is that W enters the model at two different levels. At the individual level it is absent: workers' upskilling investments and fractions are determined by the participation constraint alone. At the aggregate level it amplifies: the firm's return to raising μ is $(W + 1)$ times the per-worker upskilling sensitivity, so a more populous lower-rank group makes both viability easier to satisfy and optimal equality more aggressive. This two-level structure — individual neutrality, aggregate amplification — is the model's central mechanism.

A Complete Notation Table

B Derivation of Optimal Upskilling Fractions λ_L^* and λ_H^*

This appendix shows in full detail how the optimal upskilling fractions λ_L^* and λ_H^* in equation (30) are derived. The result does not come from setting a first-order condition equal to zero in the usual interior sense. Instead, the FOC reveals that the objective is monotone in λ_s , which forces the solution to the boundary defined by the participation constraint. We show this in six steps.

A.1 The Worker's Problem

Worker $s \in \{L, H\}$ with income M_s chooses $\lambda_s \in [0, 1]$ to maximise expected utility:

$$V_s(\lambda_s; \lambda_{-s}) = \frac{1}{2} u(b_s(\lambda_s) + \Pi_{-s}) + \frac{1}{2} u(b_s(\lambda_s) - W_s \ell_s(\lambda_s)) \quad (\text{A.1})$$

subject to the participation constraint:

$$b_s(\lambda_s) \geq \bar{b}(\mu) = \bar{b}_0 R \mu \quad (\text{A.2})$$

where the components are:

$$\kappa_s = \lambda_s M_s \quad (\text{upskilling investment}) \quad (\text{A.3})$$

$$b_s(\lambda_s) = \kappa_s + \kappa_s^\gamma - \delta_w = \lambda_s M_s + (\lambda_s M_s)^\gamma - \delta_w \quad (\text{base income after upskilling}) \quad (\text{A.4})$$

$$\ell_s(\lambda_s) = (1 - \lambda_s) M_s \quad (\text{lottery stake}) \quad (\text{A.5})$$

The prize Π_{-s} is the other worker's stake, which is fixed from worker s 's perspective since λ_{-s} is taken as given. The lottery weight is $W_L = W \geq 1$ for worker L and $W_H = 1$ for worker H .

A.2 The Unconstrained First-Order Condition

Differentiating V_s with respect to λ_s :

$$\frac{\partial V_s}{\partial \lambda_s} = \frac{1}{2} u'(b_s + \Pi_{-s}) \cdot \frac{\partial b_s}{\partial \lambda_s} + \frac{1}{2} u'(b_s - W_s \ell_s) \cdot \frac{\partial (b_s - W_s \ell_s)}{\partial \lambda_s} \quad (\text{A.6})$$

Computing each derivative in turn. Since $b_s = \lambda_s M_s + (\lambda_s M_s)^\gamma - \delta_w$:

$$\frac{\partial b_s}{\partial \lambda_s} = M_s (1 + \gamma \kappa_s^{\gamma-1}) > 0 \quad (\text{A.7})$$

Since $W_s \ell_s = W_s(1 - \lambda_s)M_s$, we have $\partial(W_s \ell_s)/\partial \lambda_s = -W_s M_s$, so:

$$\frac{\partial(b_s - W_s \ell_s)}{\partial \lambda_s} = \frac{\partial b_s}{\partial \lambda_s} + W_s M_s = M_s (1 + W_s + \gamma \kappa_s^{\gamma-1}) > 0 \quad (\text{A.8})$$

Substituting back into equation (A.6):

$$\frac{\partial V_s}{\partial \lambda_s} = \frac{M_s(1 + \gamma \kappa_s^{\gamma-1})}{2} u'(b_s + \Pi_{-s}) + \frac{M_s(1 + W_s + \gamma \kappa_s^{\gamma-1})}{2} u'(b_s - W_s \ell_s) \quad (\text{A.9})$$

Since $u'(\cdot) > 0$ everywhere, $M_s > 0$, $W_s \geq 1$, and $\gamma \kappa_s^{\gamma-1} > 0$, every term in equation (A.9) is strictly positive. Therefore:

$$\frac{\partial V_s}{\partial \lambda_s} > 0 \quad \text{for all } \lambda_s \in [0, 1) \quad (\text{A.10})$$

The objective V_s is strictly increasing in λ_s . There is no interior critical point. The unconstrained optimum is the corner $\lambda_s = 1$.

A.3 Why Both Lottery Outcomes Improve with λ_s

The monotonicity in (A.10) has a clear economic interpretation. As λ_s rises (more income to upskilling):

- The **win outcome** $M_s^{\text{win}} = b_s + \Pi_{-s}$ improves because b_s rises via equation (A.7) while Π_{-s} is fixed.
- The **lose outcome** $M_s^{\text{lose}} = b_s - W_s \ell_s$ improves by even more: b_s rises and simultaneously $W_s \ell_s = W_s(1 - \lambda_s)M_s$ falls, so both additive components move favourably.

There is no upskilling-lottery tradeoff in terms of final outcomes. Raising λ_s strictly dominates lowering it in both states of the world. The only force that prevents $\lambda_s = 1$ is the participation constraint, to which we now turn.

A.4 The Constrained KKT Conditions

Introduce a Lagrange multiplier $\xi_s \geq 0$ on the participation constraint (A.2). The Lagrangian is:

$$\mathcal{L}_s = V_s(\lambda_s; \lambda_{-s}) + \xi_s [b_s(\lambda_s) - \bar{b}] \quad (\text{A.11})$$

The KKT stationarity condition in λ_s is:

$$\frac{\partial V_s}{\partial \lambda_s} + \xi_s \frac{\partial b_s}{\partial \lambda_s} = 0 \quad (\text{A.12})$$

Substituting equations (A.9) and (A.7), and factoring out $M_s(1 + \gamma\kappa_s^{\gamma-1}) > 0$:

$$\frac{u'(b_s + \Pi_{-s}) + u'(b_s - W_s \ell_s)}{2} + \xi_s = 0 \quad (\text{A.13})$$

Solving for the multiplier:

$$\xi_s = -\frac{u'(b_s + \Pi_{-s}) + u'(b_s - W_s \ell_s)}{2} \quad (\text{A.14})$$

Since $u'(\cdot) > 0$ always, the right-hand side is strictly negative: $\xi_s < 0$. But the KKT conditions require $\xi_s \geq 0$ for a $\geq$ constraint. This is a contradiction. The contradiction means there is no feasible interior point satisfying the KKT conditions — the participation constraint must bind with equality.

A.5 The Binding Constraint Determines λ_s^*

Since V_s is strictly increasing in λ_s and the participation constraint prevents $\lambda_s = 1$, the worker pushes λ_s as high as feasible. The constraint binds at equality:

$$b_s(\lambda_s^*) = \bar{b} \iff \kappa_s^* + (\kappa_s^*)^\gamma - \delta_w = \bar{b} \iff \kappa_s^* + (\kappa_s^*)^\gamma = \bar{b} + \delta_w \quad (\text{A.15})$$

where $\kappa_s^* = \lambda_s^* M_s$. This is the equation defining $\kappa^{\min}$, the minimum upskilling investment required to reach subsistence. For the closed-form case $\gamma = 2$:

$$(\kappa^{\min})^2 + \kappa^{\min} = \bar{b} + \delta_w \implies \kappa^{\min} = \frac{-1 + \sqrt{1 + 4(\bar{b} + \delta_w)}}{2} \quad (\text{A.16})$$

Since $\kappa_s^* = \lambda_s^* M_s = \lambda_s^* R M_s^0$, the optimal upskilling fraction is:

$$\lambda_s^* = \frac{\kappa^{\min}}{R M_s^0} \quad (\text{A.17})$$

For worker L with $M_L^0 = \mu$ and worker H with $M_H^0 = 1 - \mu$:

$$\boxed{\lambda_L^* = \frac{\kappa^{\min}}{R\mu}, \quad \lambda_H^* = \frac{\kappa^{\min}}{R(1-\mu)}} \quad (\text{A.18})$$

A.6 Why $\lambda_H^* > \lambda_L^*$ Always

Since $\mu < \frac{1}{2}$ we have $R\mu < R(1 - \mu)$, so the denominator of λ_L^* is smaller than the denominator of λ_H^* :

$$\lambda_L^* = \frac{\kappa^{\min}}{R\mu} > \frac{\kappa^{\min}}{R(1-\mu)} = \lambda_H^* \quad (\text{A.19})$$

The lower-income worker always invests a larger *fraction* of income in upskilling. This is not a preference difference — both workers solve the same binding constraint — but an income-scaling effect: the same absolute floor $\kappa^{\min}$ represents a larger share of a smaller income.

A.7 Where Risk Preferences Enter

A natural question is: if both workers have the same binding constraint and the same $\kappa^{\min}$, where does the Friedman-Savage risk asymmetry between them appear? The answer is that risk preferences do not determine λ_s^* — they determine the value of the lottery participation, captured by the risk premium $\tilde{\rho}_s$ in equation (25).

Specifically, worker L participates in the lottery mechanism (at the stake level $\ell_L^* = (1 - \lambda_L^*)M_L$) because the convex curvature of utility at $b_L < m^*$ makes the gamble attractive: $\tilde{\rho}_L < 0$ means worker L receives a premium for bearing risk. Worker H participates because displacement is not fully avoidable, and their concave utility at $b_H > m^*$ means $\tilde{\rho}_H > 0$: they pay a premium to participate.

Neither premium affects the optimal λ_s^* because the risk premium correction $\tilde{\rho}_s$ is additive in V_s (see equation (29)) and does not feed back into the participation constraint. This is the key property of the linear approximation of u'' : it preserves the sign asymmetry of risk preferences while keeping the allocation problem separable from the preference problem.

A.8 Summary of the Derivation Chain

Step	Result
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A.2	$\partial V_s / \partial \lambda_s > 0$ always: no interior optimum
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A.3	Both win and lose outcomes improve with λ_s
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A.4	KKT multiplier $\xi_s < 0$: constraint must bind
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A.5	$b_s(\lambda_s^*) = \bar{b}$ gives $\kappa^{\min}$; closed form for $\gamma = 2$
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A.6	$\lambda_s^* = \kappa^{\min} / RM_s^0$; ordering $\lambda_L^* > \lambda_H^*$ from income scaling
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A.7	Risk preferences enter via $\tilde{\rho}_s$, not via λ_s^*
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The optimal λ_s^* is determined entirely by the binding participation constraint, not by equating marginal upskilling benefit to marginal lottery benefit. The FOC's role is to confirm there is no interior solution, thereby directing the solution to the constraint boundary.

Table 3: All symbols, definitions, and domains

Symbol	Definition	Domain
<i>Primitives</i>		
R	Total income to workers	$\mathbb{R}_{++}$
μ	Income share of worker L	$(0, \frac{1}{2})$
μ_0	Reference (market) income split	$(0, \frac{1}{2})$
W	Population mass ratio = $i^*/(N - i^*)$	$[1, \infty)$
$\bar{b}_0$	Subsistence coefficient; $\bar{b}(\mu) = \bar{b}_0 R \mu$	$\mathbb{R}_{++}$
δ_w	Human capital depreciation	$[0, 1)$
γ	Upskilling production power (= 2 for closed form)	$(0, 1)$
π	Firm wage share of assets	$(0, 1)$
δ_f	Firm asset depreciation	$(0, 1)$
k	Restructuring cost scale	$\mathbb{R}_{++}$
η	Restructuring cost power (= 2 for closed form)	$(1, \infty)$
c_f	Asset maintenance cost scale	$\mathbb{R}_{++}$
ξ	Asset maintenance cost power	$(1, \infty)$
α_u	Friedman-Savage curvature	$\mathbb{R}_{++}$
β_u	Friedman-Savage skewness	$\mathbb{R}_{++}$
m^*	Friedman-Savage inflection income	$\mathbb{R}_{++}$
<i>Derived constants (computed once)</i>		
κ_0	$\kappa^{\min}(\mu_0) = (-1 + \sqrt{1 + 4(\bar{b}_0 R \mu_0 + \delta_w)})/2$	$\mathbb{R}_{++}$
Ψ_0	Sensitivity $d\kappa^{\min}/d\mu$ at μ_0 : $= \bar{b}_0 R/(1 + 2\kappa_0)$	$\mathbb{R}_{++}$
Γ	Firm FOC coefficient: $= \pi(W + 1)\Psi_0/(2kR)$	$\mathbb{R}_{++}$
<i>Worker objects</i>		
M_L, M_H	Incomes = $R\mu, R(1 - \mu)$	$\mathbb{R}_{++}$
λ_s	Fraction of income to upskilling	$[0, 1]$
κ_s	Upskilling investment = $\lambda_s M_s$	$\mathbb{R}_+$
ℓ_s	Lottery stake = $(1 - \lambda_s)M_s$	$\mathbb{R}_+$
b_s	Base income = $\kappa_s + \kappa_s^\gamma - \delta_w$	$\mathbb{R}$
$\bar{b}(\mu)$	Subsistence floor = $\bar{b}_0 R \mu$	$\mathbb{R}_{++}$
$\kappa^{\min}(\mu)$	Min. upskilling; solves $\kappa + \kappa^\gamma = \bar{b} + \delta_w$	$\mathbb{R}_{++}$
$\Psi(\mu)$	Sensitivity = $d\kappa^{\min}/d\mu$	$\mathbb{R}_{++}$
λ_s^*	Optimal upskilling fraction = $\kappa^{\min}/M_s$; W -independent	$(0, 1)$
$a(\mu)$	Worker L lottery stake = $R\mu - \kappa^{\min}$	$\mathbb{R}_{++}$
$c(\mu)$	Worker H lottery stake = $R(1 - \mu) - \kappa^{\min}$	$\mathbb{R}_{++}$
$\Pi(\mu)$	Total lottery spread = $W a + c = \Lambda^*$	$\mathbb{R}_{++}$
ϵ_L^*	Net expected transfer to L = $(c - W a)/2$	$\mathbb{R}$
$K^*(\mu)$	Total upskilling = $(W + 1)\kappa^{\min}(\mu)$	$\mathbb{R}_{++}$
D_s	Lottery deviation = $\bar{b} - m^* + \epsilon_s$	$\mathbb{R}$
$\tilde{\rho}_s$	Linearly approximated risk premium	$\mathbb{R}$
$\mu^\dagger$	Social premium threshold = $m^*/(\bar{b}_0 R)$	$(0, \frac{1}{2})$
$W^\dagger$	Transfer reversal threshold = $c(\mu)/a(\mu)$	$[1, \infty)$
<i>Firm objects</i>		